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3

A HISTORY OF MARKETS PAST

The Role of Institutional Memory Failure in Financial Crises

Natalya Vinokurova

The cyclical nature of the management field's interest in the methods of anthropology and history vis-à-vis the methods of management's dominant disciplines—economics, sociology, and psychology—leads to a tendency to lump together the contributions of anthropology and history under the broader heading of qualitative methods. This tendency fosters the illusion of the two disciplines' interchangeability. This chapter uses a historical case study of financial markets to argue that the differences in methods brought by anthropologists and historians to the study reflect the different questions anthropologists and historians ask and answer. These differences point to complementarity rather than interchangeability of the contributions of anthropology and history to the study of financial markets and to the management field.

Specifically, I argue that economic crises force management as a field to develop an appreciation for fine-grained observations, rendering the methods used by anthropologists and historians particularly relevant. In this chapter, I show how historians' questions and methods differ from those of the anthropologists by comparing these questions to those of anthropologists studying financial markets and methods to those of anthropologists working with archival data. Using a historical case study to explore the connection between financial crises and the failure of institutional memory to inform the re-emergence of financial instruments, I illustrate the argument with a study of the antecedents of mortgage-backed securities (MBS), one type of securities implicated in the 2008 crisis. My study demonstrates how historical analysis can enrich the management scholars' understanding of financial crises beyond the notable contributions made by anthropologists.

Taking the 2008 financial crisis as the latest economic crisis to bring about an epistemic crisis for the field of management, I provide an overview of how anthropology and history contributed to unpacking the financial market practices in the lead-up to the 2008 financial crisis. To show the value of history's differential contribution I then lay out a detailed case study of how a historical analysis of the markets for MBS that emerged in the U.S. in the 1880s and 1900s informs our understanding of the MBS market that emerged in the 1970s and culminated in the 2008 crisis. My analysis has implications for how management scholars conceptualize the relationship between institutional memory and financial crises.

Epistemic Crises and Disciplinary Commitments

Arguably, the management field's interest in applying different research methods goes through epistemic cycles alternatively dominated by scholars closely observing phenomena and generating theory. The first phase of such an epistemic cycle entails researchers accumulating fine-grained observations of empirical reality. The contributions of anthropology and history to the management field are particularly evident during the accumulation of observations phase of the epistemic cycle. For instance, anthropologists played an important role in shaping the Hawthorne studies that laid the foundation for the development of management as a discipline (Morey and Luthans, 1987). Similarly, historical analysis of the emergence of the M-form corporation (Chandler, 1962) drew on detailed archival records to inform management theory. These contributions leverage anthropologists' and historians' interest in understanding the contexts that shape human behavior, be they aspects of a society's culture or evolution of its institutions.

In the second phase of the epistemic cycle, after enough data are accumulated, management field's interest in the case particulars wanes and the careful images painstakingly drawn by the original researchers are replaced first by outlines and then by outlines of the outlines, i.e., the stylized "facts."¹ This theory-generation phase of the cycle draws heavily on economics, sociology and psychology, relying on tools of what Abbott (2004) termed "standard causal analysis." These dominant disciplines seek to sketch out generic patterns of human behavior invariant to space, time, and cultural context. An example of the first two phases of such an epistemic cycle is the evolution in the disciplinary orientation at university business schools from closely observing the phenomena and collecting facts on individual cases to generating theories that draw on the dominant disciplines (Khurana, 2010).

Epistemic crises—significant events for which the theories fail to account—restart the epistemic cycles. By demonstrating the limitations of assumptions inherent in the prevailing theories, epistemic crises disrupt the theories' dominance, generating greater interest in the context. The evidence of the theories' limitations points to the need to understand the individuals' cultural beliefs and histories of the institutions that shape them, stimulating management scholars' interest in the study of anthropology and history, and thus the start of a new cycle of inquiry. After the

crisis, this accumulation of observations gives rise to theories that abstract away from the particulars until the next crisis restarts the cycle.²

Economic crises play an important role in the epistemic cycles for management studies. As an applied field of inquiry, management is committed to the study of reality—a commitment that turns economic crises into epistemic crises for the field. The aftermath of economic crises—events that force the field to recognize the limitations of existing theories—brings to the fore the questions overlooked by the field's dominant disciplines. This recognition points to the inadequacy of the prevalent theories, and the need to collect more data and critically evaluate the interpretations of the data already collected.

In the aftermath of economic crises, management scholars are called upon to account for the gap between what the theory says and what managers do. Offering this account is central to the management field's contribution to the broader body of knowledge. Anthropology and history shape this contribution at moments of crisis by collecting and making sense of observations that help explain this gap.

From Proximate to Distant Antecedents of the 2008 Financial Crisis

The 2008 mortgage crisis represents an opportunity for sensemaking brought about by the limitations of both academic (Krugman, 2009) and practitioner theories (MacKenzie, 2011) in accounting for the event. The massive foreclosures on residential real estate in the United States affected millions of families, bringing about bankruptcies of major financial firms and protracted unemployment. The need to make sense of the 2008 crisis produced an interest in understanding the work practices of financial market participants, energizing the social studies of finance as an emergent discipline (MacKenzie and Millo, 2003; Poon, 2009). This analytical lens built on ethnographic accounts of the financial services that emerged in the work of academics (Abolafia, 1996; Ho, 2009; Knorr Cetina and Bruegger, 2002; Zaloom, 2006), journalists trained in anthropology (Tett, 2009), and practitioners (Lewis, 1990, 2010).

In exploring the evolution of the securities implicated in the financial crisis, scholars taking a historical lens focused on the role played by innovations in security design and the practices surrounding such innovations (Funk and Hirschman, 2014; MacKenzie, 2011; Vinokurova, 2018, 2019a). These accounts leveraged the relative recency of the 2008 crisis to excavate the immediate antecedents of the securities in question, supplementing archival sources with practitioner interviews. However, they did not take advantage of the wider lens afforded by the historical method in interrogating the more distal ancestors of the securities in question. In this chapter, I contribute to filling this gap by analyzing the connections between mortgage-backed securities (MBS), which were one type of securities widely used in the lead-up to the 2008 crisis and the prior U.S. MBS markets that emerged in the 1880s and 1900s. In so doing, I highlight the differential contribution of history to the study of financial markets.

Learning from Prior Markets

In this chapter, I consider the connections between the 1880s, 1900s, and 1970s MBS markets in the United States. The emergence and re-emergence of these markets offer insights into the role played by institutional memory in the MBS market development. The U.S. MBS markets are of particular interest because of the discontinuities in their operations. Other countries' forays into mortgage securitization exhibit evolutionary change—the markets in Germany, France, and Denmark span centuries.³ By contrast, the U.S. markets emerge, disappear after a crisis, and reemerge.

My analysis focuses on securities offered to U.S. investors backed by residential real estate. This focus rules out the securities of U.S. mortgage companies created in the 1870s that targeted European investors (Brewer, 1976), the shares of commercial mortgages offered by bond houses in the 1920s (Snowden, 1995), and the farm mortgage lending system created by the U.S. government in 1915 (Sparks, 1932). The three markets that I consider in this chapter are for the debentures backed by western mortgages issued in the 1880s and 1890s, the mortgage participation certificates issued between the 1900s and 1930s, and different types of MBS issued in the most recent market from the 1970s through 2008.

Focus on Financial Instruments

Economic historians have argued that the connection between the MBS market meltdowns of the 1890s and 1920s is that the U.S. mortgage lenders missed the important institutional lessons of the European securitization experience. For instance, Snowden (1995) contrasts the lack of a centralized regulator for mortgage lending and the issuance of MBS in the United States with the highly centralized regulation of mortgage banking in Europe. Other scholars have similarly pointed to private market competition as contributing to the U.S. MBS crises (Simkovic, 2013). These accounts have stopped short of comparing and contrasting the structures of the securities issued across the three markets.

Comparing and contrasting the security structures across the three markets is important because such analysis can document the institutional memory (or lack thereof) in the U.S. MBS markets and the financial system more broadly. Since all three MBS markets ended in crises, comparing the security structures could inform debates about whether and what market participants learn from financial crises. Persistence of the security structures would suggest that MBS market participants failed to learn not just from the overseas experience with mortgage securitization, but also from the experience of their U.S. predecessors. Furthermore, evidence of persistence in mortgage securitization practices would lend further credence to the argument that the regulators shape the scope of action available to the financial markets participants (Funk and Hirschman, 2014).

Accounts of the MBS market development in the lead-up to the 2008 crisis have focused on recent past, thus leveraging the complementarities in the methods of anthropologists and historians. Taking a longer historical perspective allows me to highlight the differential contribution of history.

Sources and Methods

In tracing the history of MBS across the three markets, I collected more than 500 printed primary sources. The primary sources consisted of MBS issuers' advertisements in newspapers and magazines,⁴ the discussion of the mortgages and MBS in the trade press, the reports of the authorities supervising the markets,⁵ as well as the market post-mortem reports of the relevant investigating bodies.

Each source offered its own specific lens on MBS. The advertisements placed by the MBS issuers offered a repository of claims made about the securities including the structure of the securities, the rate of interest, and the targeted audience. The persistence of advertising as a medium provides an opportunity to compare and contrast the rhetoric of the MBS issuers across the three markets. The trade press discussions of mortgages and MBS reflect the market participants' views on MBS and the evolution of these views across the markets. The regulator reports offer insights into regulatory decision-making with respect to the MBS issuer activities. The post-mortem reports of the three markets contain the lessons learned by both the private actors and the policy makers about the markets in question. I supplemented these primary materials with academic accounts of the three markets, including both contemporaneous accounts of the MBS issuers' activities and the analyses of the markets following each market's collapse.⁶

My analytical lens reflects three key aspects of a historical approach to collecting and analyzing archival sources (Kipping, Wadhwani, and Bucheli, 2014): source criticism, triangulation, and hermeneutics. Source criticism entails the need to establish both the authenticity (time and source of production) and validity (the reliability of the informants producing the documents) of the sources. Triangulation suggests that drawing on multiple written sources can help the historian establish a more accurate understanding of the relevant context, overcoming the biases in the individual sources. Finally, hermeneutics requires that an understanding of how the production of the source material fit into the historical context informs the historian's interpretation of the sources.

This approach has important similarities and differences to how an anthropologist might approach analyzing archival sources. For instance, in their analysis of how writing affects distributed collaboration, Fayard and Metiu (2014) analyze two sets of letters. Their analysis focuses on iteratively reading and coding the archival materials, using extant theory to inform their coding, and presenting their findings in relation to the relevant theories. A historian approaching the same set of materials would start by iteratively reading the primary source documents (in this case the letters) and contemporaneous secondary or tertiary sources, including academic accounts. The historian's "coding" of the letters would then be informed by their relationship to other archival sources and other historians' accounts of the period. The historian would then present the findings in relation to what is known about the period in question.

In presenting my findings, I begin with an overview of the structures of MBS to provide the reader with a basis for comparing the securities in the three markets. I then offer brief descriptions of the three markets in turn. Following these

descriptions, I compare and contrast the structures of the securities across the three markets. For each market, I consider the lessons the market participants drew from their experience and investigate the extent to which the subsequent MBS market participants learned from the experience of their predecessors. I conclude my analysis by discussing how this comparison informs our understanding of the relationship between institutional memory and financial crises.

MBS across Time

Before comparing the security structures across the three markets, I offer an overview of MBS as a financial instrument. As illustrated in Figure 3.1, the starting point in MBS issuance is the assembly of individual mortgages into a pool. The MBS issuers then use the mortgages in the pool as the securities' collateral using the cash flows from the mortgages in the collateral to pay the securities' investors.

Depending on the structure of an individual issue, the MBS may represent either a debt obligation of the issuer or partial ownership of a pool of mortgages. In the former case, for the purposes of the issuer's balance sheet, the mortgage pool appears as an asset and the securities appear as a liability of the issuer. In the latter case, the issuer sets aside the mortgages by either depositing them with a trustee or setting up a separate entity to issue the securities and selling the mortgages to that entity.

There are three key assumptions underlying the issuance of MBS. First is that the quality of the mortgages entering the mortgage pool is comparable to the quality of the mortgages sold individually. This assumption undergirds the comparability of yields and risk profiles of MBS to those of the individual mortgages. Second, is that the quality of underlying mortgages does not change within markets. This assumption allows for the comparability of yields of MBS issued at different points in time. Third is that the standardization of inputs (mortgages) enables the standardization of outputs (securities).

Insight into MBS market development in the U.S. is constrained by the lax regulatory framework and fragmentation of the supervisory authority governing

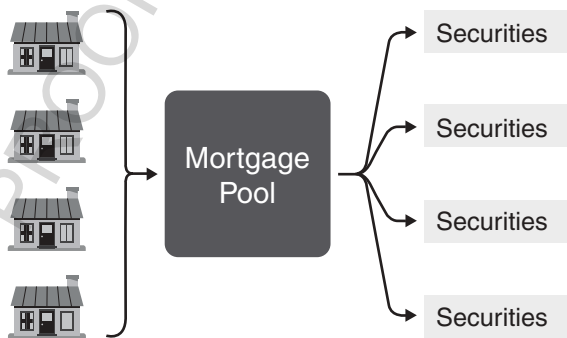


Figure 3.1 Structure of mortgage-backed securities.

mortgage lending and MBS issuance. The laxity encompasses an absence of uniform standards for mortgage underwriting and recording, lack of uniform licensing standard for mortgage underwriters and MBS issuers, as well as a dearth of requirements for transparency into the securities collateral. The fragmentation stems from a multitude of regulators governing different aspects of mortgage lenders' activities. These features of the regulatory framework make it difficult to judge how well the assumptions implicit in the MBS structure compare to reality.

MBS: 1880s–1890s

In the mid-1800s, investors from the east coast of the United States were an important source of mortgage financing for farm and city property in Midwestern and Western states. In the 1880s, the western mortgage companies started financing the mortgages they originated by issuing debentures to cater to investors who lacked sufficient capital to buy individual mortgage loans. These securities gave investors the option to invest in western mortgages by buying obligations of the mortgage lenders, collateralized by the mortgage loans the companies deposited with a trustee. The mortgage lenders collected borrower payments, passed them on to investors, and offered a guarantee of principal and/or interest repayment on the mortgages they originated. The guarantees allowed investors to forgo the expense of researching the quality of individual mortgage loans.

Figure 3.2 displays one of the earliest advertisements for western mortgage debentures published in *Christian Union*—a nondenominational general interest publication. The advertisement was accompanied by text explaining the security's features:

These debentures are obligations of the company, who are responsible for the prompt payment of both interest and principal. They are secured by first mortgages on improved farms, and for each series of \$50,000 an equal amount of these mortgages are on deposit with the Metropolitan Trust Co. of New York, and do not represent in the aggregate more than 50% of the value of the property. The Treasurer of the company, William G. Clapp, 7 Nassau Street, New York, will send a pamphlet explaining more fully their debenture system, or give any further information that may be desired.

(Christian Union, 1886, p. 30)

The advertisement and the accompanying text suggest that in structuring its debentures, the Fidelity Loan and Trust Company of Storm Lake, Iowa had opted for issuing the securities as its debt obligations, while at the same time depositing the mortgages with a trustee to create the appearance of separation between the issuer and the securities' collateral. The securities' description also offers assurances of the quality of mortgages—the amount of the loan was not to exceed 50%. The description of the mortgaged properties as “improved farms” in the ad is meant to reassure the prospective investors that the mortgaged property in the securities' collateral

**ANET 6%
INCOME**

THE
Fidelity Loan and Trust Co.,
OF STORM LAKE, IOWA,

Offers to investors, at par and interest, its bonds bearing 6 per cent. (semi-annual) interest secured by mortgages on improved real estate, deposited with the
Metropolitan Trust Co. of New York,
TRUSTEE.

THESE BONDS (\$1,000) ARE ISSUED IN SERIES OF \$70,000, EACH ISSUE HAVING AN INDEPENDENT MORTGAGE SECURITY REPRESENTING OVER \$140,000 OF PROPERTY, AND ARE RECOMMENDED AS A SUPERIOR INVESTMENT FOR TRUST AND SIMILAR FUNDS FOR WHICH SAFETY IS THE FIRST REQUISITE.

Full information and bonds for sale at offices of
WM. G. CLAPP, Treas.,
7 Nassau St., N. Y.
JOHN PATON & CO.,
Financial Agents,
52 William St., N. Y.

Figure 3.2 An advertisement of Western mortgage debentures offered by Fidelity Loan & Trust Co. of Storm Lake, Iowa.

Source: *Christian Union*, Advertisement 15, October 21, 1886, p. 29.

did not include vacant land. Of note in the text above is the idiosyncratic nature of the securities in question—the readers are advised to contact the company for a description of the company-specific debenture issuance system.

In addition to the above description of the securities, the text accompanying the ad provides some background on factors motivating the issuance of the securities:

The immense and increasing demand in the agricultural districts of the West for Eastern capital to improve and stock newly purchased farms has led to great improvement in the method of negotiating Western farm mortgages. The debentures issued by the Fidelity Loan & Trust Co. of Storm Lake, Iowa, advertised on another page, belong to a system for which many advantages are claimed, and is worthy of the careful investigation of any one interested in this class of investments.

(*Christian Union*, 1886, p. 30)

The above text suggests that the farmers' demand for capital drove the issuance of the new securities. The quote also frames the debentures as a better way to invest in western mortgages.

The operation of western mortgage companies issuing debentures proceeded with no regulatory oversight until the early 1890s when bank supervisors in the

eastern states in which the companies recruited investors started issuing licenses conditional on examining firms' finances (Bogue, 1955). The regulators were careful to explain that the licensing for operation did not constitute endorsement:

While farm mortgages, or debenture bonds based upon them, are undoubtedly safe investments when the business is *honestly* and *capably* managed, we wish it to be understood that the mere appearance of the statement of any company in this report does not, of necessity, carry with it the endorsement of the Commissioners, as for want of time and the fact that many companies were late in qualifying, some of them have not yet been examined. Neither statute law nor State supervision can give to the investor absolute protection; he must, to a great extent, rely on his own judgment and caution.

(Griswold and Goodrich, 1890, *Report of the Bank Commissioners of the State of Connecticut*, p. 17)

In keeping with the regulators' caveats in the above quote, in their advertisements, the mortgage companies sought to project the image of care in their lending. These claims included lending to good borrowers, on perfect titles, in amounts not exceeding 40% of the property value (e.g., *Philadelphia Inquirer*, 1886). While the regulators were careful to avoid the appearance of endorsing the companies, the companies advertised their products as comparable in safety to government bonds (see for instance the Equitable Mortgage Company ad in the *Christian Union*, 1886, p. 29).

A piece of evidence consistent with the claims that the debentures were a safer means of investing in western mortgages, the companies advertised a lower interest rate for their debentures than for individual mortgages they sold, as seen in the Farmers' Trust Company ad in Figure 3.3.

The 1% difference in annual interest rates between the debentures and the individual mortgages in Figure 3.3 indicates greater safety claimed for the debentures.

Farmers' Trust Co.
Chicago Office, 111 Dearborn-st.
6 Per Cent Savings Bonds.
6 Per Cent Registered Debentures.
7 Per Cent Western Farm Mortgages.
Denominations to suit Investors or Savings Depositors.

Figure 3.3 Advertisement of western mortgages and debentures offered for sale by the Farmers' Trust Co.

Source: *Chicago Daily Tribune*, 1887. "Display Ad 4", July 14, p. 12.

In appealing to investors, MBS issuers advertised the risk-diversification benefits of investing in multiple mortgages as offering protection for the investment's performance compared to buying individual mortgages. For example:

The advantages of the debenture system are that the investor is not compelled to stand or fall with one mortgage or one piece of real estate. Each debenture bond is, in a sense, insured by all the rest of the series.

(Gleed, 1890, p. 104)

Eastern regulators cited diversification as a rationale for licensing companies whose capital was less than one-sixth of the mortgages they guaranteed:

Forty-three companies, organized under the laws of other States or of territories, have placed themselves under the supervision of the commissioners. Sworn reports, showing the condition of such companies on the first day of October last, will be found following the reports of the home companies. The capital stock represented by these companies is \$9,171,771.26, and they have an aggregate liability for debenture bonds of \$9,851,568.86, with a still further liability of \$45,342,413.43, for outstanding guaranteed loans negotiated and sold by them. These liabilities amount to more than six times the amount of capital represented; but, in justice to the companies, it should be stated that **they consider the risk to be governed by the law of average and in the nature of an insurance** [emphasis added].

(Landers and Noble, 1888, Report of the Bank Commissioners of the State of Connecticut, p. 15)

The benefits of the risk diversification that the companies claimed hinged on the assumption that the quality of the mortgages in the debentures' collateral was comparable to the quality of mortgage loans sold individually—an assumption that was at odds with the practice of the securities' issuers.

Indeed, market observers suggested that in the absence of close regulatory supervision, mortgage companies selected lower quality collateral for the securities compared to the individual loans they sold to other investors:

The chief objection to what is called the debenture system is that **companies are likely to secure their debentures by a poorer class of mortgages**. The trustee never vouches for the character of the securities upon which debenture bonds are based. He only certifies the face value. Bad securities taken for large commissions are likely therefore to be put into the hands of the debenture trustee. There is nobody to inspect or to criticise [emphasis added].

(Gleed, 1890, p. 96)

As this quote suggests, the trustees who held the debentures' collateral played a passive role in the issuance of securities—attesting to the face value of the mortgages placed in the trust rather than the quality of the mortgages.

Eastern bank examiners urged investors to pay attention to the securities' collateral:

No supervision, however, can protect the investor unless he uses good judgment in the selection of the company with which he deals and carefully scrutinizes each investment, if it be a loan, to study its locality and all its conditions from the memoranda furnished for that purpose, and if it be a debenture bond, to be sure that it states clearly the security [collateral] on which it is based.

(Landers and Noble, 1889, Report of the Bank Commissioners of the State of Connecticut, p. 19)

The regulators' guidance was particularly salient because even reputable lenders like J.B. Watkins in Kansas used the mortgage loans they could not sell individually to back the debentures. These loans had shorter terms, smaller amounts, and higher risk compared to the mortgages that were sold as individual loans (Bogue, 1955). Further compounding the problem of collateral quality was the fact that in addition to selecting lower quality mortgages to begin with, the security issuers could also replace the loans held by the trustees on the investors' behalf at the issuers' discretion.

In the lead-up to the panic of 1893, several developments contributed to the collapse of the market for western mortgages. One of these was overlending. The abundance of capital led to the lowering of mortgage interest rates as well as to unsafe lending practices. As the below quote suggests, the easy availability of capital staved off foreclosures by turning delinquent loans into new mortgages:

A Western farm mortgage on which over a year's interest was in default was sent from the East to a Chicago firm for collection, and by it sent to a Western correspondent, who reported the borrower "badly on the down grade." A few days later word came that it would be paid, and a little later full payment was received. Being a little curious on the subject, further inquiry was made, and a reply came that a Western mortgage company had made delinquent a new loan for 40% more than the old one—adding, for which a round commission was paid.

(New York Evangelist, 1886, p. 7)

As this quote suggests, the easy availability of capital contributed to a reduction in the quality of the loans in the debentures' collateral. The crop failures of the late 1880s resulted in defaults of many western mortgages. The concerns about farm mortgage performance in light of crop failure curtailed the flow of capital to western mortgages, reducing opportunities for refinancing. By the mid-1890s, most of the mortgage companies failed, unable to honor their guarantees due to insufficient capitalization.

MBS: 1900–1930s

As the issuance of debentures backed by western mortgages wound to a close in the 1890s, securitization reemerged as an important financing source for urban mortgage lending. Specifically, the companies originally formed to insure the validity of property titles in American cities expanded their businesses into insuring the repayment of mortgages and mortgage lending. Like the western mortgage companies before them, the title insurance companies offered investors both individual mortgage loans and securities backed by mortgage loans. However, instead of issuing debentures that constituted debt obligations of the western mortgage companies, they sold mortgage participation certificates—securities that represented partial ownership of the mortgage loans these companies originated.

Figure 3.4 offers three advertisements representative of the claims made by the certificates' issuers.

The claims in the mortgage certificate ad in Figure 3.4a echoed the claims made by the western mortgage companies in the 1800s. The issuers advertised the mortgage certificates as an improvement over prior methods of investing in mortgages, promising the mortgage holders a stream of income without having to concern themselves with investigating the details of mortgage lending practices.

An Improvement

Do you know the latest development of the guaranteed mortgage—The First Mortgage Certificate?

It is a share in a mortgage on a definite piece of property. You have all the protection of the guaranteed mortgage but there are no bulky papers to care for—we do that for you. You hold the certificate—we do all the rest and your interest reaches you each six months.

Just now they pay 5% and you can get them in any amount, odd or even, small or large. They are due in from three to five years.

Try them the first time you have money to invest.

We have guaranteed \$692,000,000 in the past 27 years and no investor has ever lost a dollar.

BOND & MORTGAGE GUARANTEE CO.

Capital and Surplus, \$10,000,000

176 Broadway, New York
175 Remsen St., 196 Montague St., Bklyn
350 Fulton St., Jamaica
67 Jackson Ave., Long Island City

Can You Save?

\$3.72 a Month?

ON payments as low as \$3.72 a month you can buy a 5½% Guaranteed First Mortgage Certificate—and each payment will draw full 5½% interest.

These Certificates are issued under rigid State regulations. They make you actual part owner in absolute First Mortgages on selected real estate.

These mortgages are never taken for over 50% of value and the appraisals are reported by the State. These mortgages are also covered by Policies of Mortgage Insurance backed by a \$2,500,000 Capital.

Certificates in \$100 multiples. Ask for Brochure "B."

UNDER STATE SUPERVISION

MORTGAGE GUARANTEE CO.

Fully-Paid Capital \$2,500,000
626 So. Spring

Guaranteed-Certified

It makes a great difference when you invest your money whether you, yourself, take the risk of ever seeing it again or whether its return is guaranteed by someone able to make good his guarantee.

When you buy a bond you do not get such a guarantee. The risk is yours. When you buy a Guaranteed First Mortgage Certificate you are certified that the risk is wholly on the Bond & Mortgage Guarantee Company.

During the past twenty-eight years more than \$750,000,000 of this class of security has been bought in reliance on such guarantee, without the loss of one dollar of principal to an investor or one day's delay in payment of interest.

Bond & Mortgage Guarantee Co.

176 Broadway, New York City
175 Remsen Street, Brooklyn

Figure 3.4 Three advertisements of guaranteed mortgage certificates.

Source: New York Times, Display Ad 5, Jan. 17, 1916, p. 5; Los Angeles Times, Display Ad 47, Jun. 25, 1915, p. 118; New York Times, Display Ad 94, Jan. 11, 1921, p. 19.

As with the farm debentures in the 1880s, the issuers of the mortgage certificates targeted individual investors and, more specifically, smaller individual investors. The advertisements in Figure 3.4 show a variety of claims made by the MBS issuers to assure investors of the securities' safety. The ad in Figure 3.4a cites the investors' prior experience with guaranteed mortgages suggesting that no investor ever lost money on the securities. Of particular interest in Figure 3.4b are the appeals to state regulation and supervision as a means to assure the investors of the safety of the securities. The Mortgage Guarantee Company claimed to issue the securities both "under rigid state regulations" and "under state supervision." These claims created an illusion of government supervision if not endorsement of the securities.

As indicated in Figure 3.4c, the issuers of guaranteed mortgage certificates emphasized the value of the guarantee, arguing that the guarantee made their securities superior to other securities, lacking such guarantees, e.g., the unguaranteed [real estate] bonds. A 1923 *Wall Street Journal* article summarized the discussion of the relative merits of the guaranteed mortgage certificates and the unguaranteed real estate bonds, issued to help finance large commercial building, as the following dialogue between the issuers of the two security types shows. The mortgage insurance companies emphasized the role of guarantees and the high rates of interest offered by their products: "An Investor should choose his bond by noting whether the mortgage company will put all its resources as a guarantee behind the issue it sells, and by the rate of interest which these bonds bear" (*Wall Street Journal*, 1923, p. 5). Representing the views of the real estate bond issuers, the same article asked (p. 5): "In the final analysis, what does your guarantee amount to? Considering your guaranteed mortgages to be contingent liabilities, are your assets sufficient to meet them?" The skepticism of the guarantees expressed in the article proved to be well founded.

While the firms issuing the mortgage certificates claimed to focus on residential mortgages, their mortgage lending activities also included higher-risk lending on commercial real estate and vacant land (Alger, 1934). The mortgage participation certificates issued by the mortgage insurance companies in the 1900s were similar to debentures of the 1880s in that the mortgage insurance companies also designated certain mortgages on their books as backing specific mortgage certificates. However, instead of depositing the documentation for the mortgages backing these certificates with independent trustees, the certificate issuers used their own subsidiaries as both trustees for the mortgage documentation and depositories for collecting the interest and principal payments from the mortgage borrowers (Alger, 1934).

As with the debentures of the 1880s, the MBS issuers exercised control over which mortgages would be included in the certificates' collateral. This control allowed the issuers to select lower quality mortgages (e.g., mortgages on vacant land or commercial properties) as collateral for the certificates. Using practices that echoed those of J.B. Watkins in the 1880s, one criterion that marked mortgages for inclusion in the certificates' collateral was whether investors interested in buying

individual mortgages rejected the loans in question. In the words of a vice-president of Long Island Title Guarantee Company:

When a mortgage couldn't be sold or I had certain objections given me on a certain mortgage, why then the next time a certificate request came in I would check a certificate against it, or if there were any vacant land mortgages, they would undoubtedly be put in certificates.

(Alger, 1934, p. 95)

Furthermore, after issuing the securities, the issuers removed higher-quality mortgages from the securities' collateral and replaced them with lower-quality ones. Such replacement would occur when other investors expressed interest in buying the specific loans or when the guaranteeing firms needed to post high-quality collateral to borrow money from the Reconstruction Finance Corporation (RFC)—a federal entity created to provide relief from the Great Depression (Alger, 1934, pp. 118–120).

As with the western mortgage companies of the 1880s, the mortgage insurance companies operated with few regulatory constraints. In theory, the New York State Department of Insurance had regulatory authority over the title insurance companies, issuing guaranteed mortgage certificates in New York. In practice, the department lacked the resources necessary to offer effective supervision and even when the department's inspectors identified problems with the companies' operations, they lacked the authority to order corrective action.

The real estate downturn of the Great Depression resulted in bankruptcies of the vast majority of the mortgage insurance companies and losses for the certificate holders. In a situation reminiscent of the 1880s market, the availability of mortgage financing before the crisis led to a relaxation of lending standards and overlending. As in the 1880s, the amount of mortgages the companies guaranteed dwarfed the companies' capital. Exacerbating the problem of insufficient reserves, the MBS issuers' failure to diversify their investments away from real estate hampered their ability to honor the guarantees.

In both the 1880s and the 1900s, MBS issuers invested the reserves designated for honoring mortgage repayment guarantees in real estate. The lack of diversification away from real estate made the issuers' guarantees and solvency vulnerable to downturns in real estate prices. In both the 1890s market crisis and the Great Depression, as the real estate prices dropped, the companies' assets invested in mortgages depreciated rapidly, thus rendering the companies unable to honor the guarantees embedded in the securities. This combination of issuer insolvency and lower mortgage quality in the securities' collateral translated into losses for MBS investors.

In the aftermath of the Great Depression, the federal government responded to a drop in residential mortgage lending activity by creating a host of government institutions to stabilize and inject liquidity into the mortgage market. These institutions included the Home Owners Loan Corporation (HOLC)—an entity that bought and restructured defaulted loans; the Federal Housing Administration (FHA)—an

agency that offered mortgage insurance backed by the federal government; and the Federal National Mortgage Association (FNMA or Fannie Mae)—an agency tasked with buying mortgages from mortgage lenders, thus creating a secondary market for mortgages.

MBS: 1970s Onward

The third MBS market grew out of government efforts to shift the mortgage-market stabilization efforts to the private sector. To this end, in 1968 the U.S. Congress authorized the issuance of MBS bearing the full-faith and credit guarantees of the federal government. The 1968 Housing and Urban Development Act authorized the Government National Mortgage Association (GNMA or Ginnie Mae—an agency of the U.S. government) to guarantee the repayment of MBS collateralized by loans, the repayment of which was either insured by the Federal Housing Administration (FHA) or backed by the Veterans Affairs (VA) administration. The Act left open the exact structure of the securities to be issued: “Securities issued under this subsection may be in the form of debt obligations or trust certificates of beneficial interest, or both” (Housing and Urban Development Act of 1968, [82 STAT], Section 804).⁷

The securities issued under the Act took two forms—mortgage-backed bonds (MBBs), which represented a debt obligation of their issuers, thus resembling the debentures of the 1880s, and pass-through certificates, which represented partial ownership in a pool of mortgages sold by the issuer, thus resembling the mortgage participation certificates of the 1900s. Over time, as both types of securities struggled to attract investors, the line between the debt obligation status of the MBBs and sale of assets status of pass-through certificates became increasingly blurred.

In 1975, savings and loan associations started issuing MBBs not backed by the government guarantees. Purchasing these securities required faith in the credit quality of the issuers. To develop such faith, the MBS issuers posted collateral that exceeded the face value of securities and agreed to replenish the mortgage collateral as borrowers repaid the underlying mortgages. These efforts at collateral management did little to assuage investor concerns about the link between the mortgage collateral and the issuers’ balance sheet. In the first two MBS markets, control over the securities’ collateral allowed the issuers to replace the higher-quality loans with lower-quality loans, thus diluting the quality of the collateral. In the third MBS market, investors worried that in the event of the issuers’ bankruptcy other creditors would claim the cash flows from the high-quality collateral backing the securities.

To address this concern, MBBs issuers separated the securities’ collateral from their balance sheets in the next generation of MBBs by setting up shell corporations—separate legal entities to issue the securities. The savings and loan associations transferred the mortgage collateral to the shell corporations; the shell corporations issued the securities and transferred the proceeds of such issuance back to the savings and loan associations. While this move protected the investors from other claimants on the mortgage collateral cash flows, it fully transferred the risks associated with

the mortgage collateral to the investors. Thus, effectively, the securities still labeled bonds stopped being the debt obligations of the original mortgage lenders, becoming closer in form to the 1970s pass-through certificates and the mortgage participation certificates of the 1900s.

The pass-through certificates issued in the 1970s represented shares in pools of mortgages and derived their name from passing through the principal and interest payments on mortgages to the investors. The mortgage loans backing the certificates in the 1970s differed from their pre-Depression predecessors. Specifically, the 1970s mortgages featured longer maturities, no prepayment penalties, and monthly payments of principal and interest rather than the semi-annual interest payments and a single payment of principal made by pre-Depression borrowers.⁸ This combination of features made the purchase of individual mortgage loans less attractive to both individual and institutional investors.⁹ The changes in the mortgage features meant that the certificates conveyed to the investors a stream of monthly payments of principal and interest in unpredictable amounts.

Responding to the concerns of the institutional investors used to processing fixed semi-annual bond interest payments, in 1975 Federal Home Loan Mortgage Corporation (FHLMC or Freddie Mac, a quasi-government agency created to develop the MBS market) introduced pay-through certificates—securities that represented shares in a pool of mortgage assets, but made semi-annual interest and annual principal payments. Freddie Mac's proprietary name for the pay-through certificates was Guaranteed Mortgage Certificates—a name that matched the generic name of the mortgage participation certificates of the 1900s. Echoing the practices of the MBS issuers in the first two MBS markets, Freddie Mac claimed that its securities represented both improvement and novelty, as the ad in Figure 3.5 indicates.

The 1970s pay- and pass-through certificates were structured using inheritance law to avoid the double taxation of the interest and principal received from the mortgage collateral backing the securities. The double taxation would arise if the borrowers' payments counted as taxable income for both the trustee holding the mortgages and the investors holding the certificates. To avoid this situation, the certificates' issuers deposited the mortgages backing the securities into passive trusts—an arrangement that precluded the certificate issuers from changing not just the composition of the collateral, but also the distribution of the cash flows associated with the securities. This structure safeguarded the collateral from the issuers' or the trustees' interference; however, it also passed on the inconveniences associated with investing in mortgages directly to the certificates' investors. These inconveniences included the inability to predict the timing of the mortgages' repayment, making it difficult for the investors to anticipate both the amount of time the securities would remain outstanding and the yield on the securities.¹⁰

MBS issuers worked around the unpredictability problem by directing the principal payments received by the trustee at different times to different classes (or tranches) of investors. Unable to obtain the desired tax treatment for the resultant securities when issued as pay-through certificates, Freddie Mac opted to issue them as debt obligations—with the security name changing from guaranteed mortgage

Six questions from institutional investors and The Answer

The Federal Home Loan Mortgage Corporation
announces the Guaranteed Mortgage Certificate. It's a new kind
of mortgage security. From a pool of residential mortgages.
It's for the institutional investor.

Question No. 1: What institutions will be interested?

The Answer: Pension Funds. Trust Funds. Banks.
Insurance Companies. And Thrift In-
stitutions.

Question No. 2: What will make the institutions inter-
ested?

The Answer: Several things.

- 1) The yield is competitive.
- 2) The return of principal provides cash flow.
- 3) You get principal returned once a year. This makes it easy to handle.
- 4) You may hold it for 15 years. Or, depending on the market, you may opt to hold it for the life of the mortgages which is 30 years or less. *It's your option.*

Question No. 3: How safe are Mortgage Certificates?

The Answer: The Federal Home Loan Mortgage Corporation unconditionally guarantees principal and interest on the underlying mortgages.

It also unconditionally warrants the timely payment of interest and minimum annual return of principal on the Mortgage Certificates regardless of the status of any individual mortgage loan.

And, the underlying mortgages are of high quality. They represent large numbers of conventional residential mortgages throughout the United States.

Question No. 4: How does the Mortgage Certificate differ from other mortgage instruments?

The Answer: It pays interest semiannually. It returns principal once a year. You can sell it back to the Federal Home Loan Mortgage Corporation in 15 years, at your option. If held 15 years, the maximum average life of the investment is 10.5 years. And it's been tailored for the institutional investor.

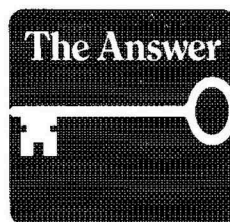
Question No. 5: Will it help the economy?

The Answer: Sure. The sale of Mortgage Certificates will provide additional money for mortgages and help those industries heavily dependent on housing.

Question No. 6: Is it "The Answer" to all my institutional money problems?

The Answer: Certainly not. But it was designed for you. It didn't just occur. It was engineered for those institutions that can profitably use cash flow. And the once-a-year principal payment is easy to administer.

• For more information please call your Securities Dealer or send for our brochure and prospectus.



Federal Home Loan Mortgage Corporation
311 First Street, N.W.
Washington, D.C. 20001

Figure 3.5 Advertisement of a Guaranteed Mortgage Certificate issued by the Federal Home Loan Mortgage Corporation (FHLMC or Freddie Mac).

Source: *Wall Street Journal*, Display Ad 72, February 5, 1975, p. 17.

certificates to collateralized mortgage obligations (CMOs). The issuance of CMOs in 1983 blurred the debt obligation line between MBBs and pass-through certificates. The Tax Reform Act of 1986 created Real Estate Mortgage Investment Conduits—a type of MBS that offered investors the tax benefits of investing in debt securities independently of whether the securities' issuer treated the issuance as a debt obligation or as a sale of assets. Effectively, the Act severed the connection between the issuer's accounting treatment of the securities and the tax treatment of the investors (Vinokurova, 2019b).

The issuance of Collateralized Mortgage Obligation (CMO)—a security accepted by the bond investors (Vinokurova, 2018)—was a culmination of a process in which over time, the more bond-like MBS retained the bond label while acquiring more mortgage features. The MBS issuers transferred the tranching tools introduced to increase the predictability of the securities' cash flows in the 1980s to default risk in the 1990s, contributing to the development of a market for non-government insured MBS. Thus, in the third market, the application of private risk management tools that leveraged diversification of risk replaced government guarantees. This market crashed in 2008 because of overlending, which rendered the tranching tools ineffective at protecting investors from default risk (Vinokurova, 2019a).

Similarities across Three MBS Markets

To explore the role played by the failure of institutional memory in the development of the three markets, I will focus my analysis on three important features of the markets—the security design with respect to the securities' debt obligation status, the guarantees offered by the securities' issuers, and the failure to recognize the role played by systemic risks in unraveling the three markets.

Did Securities Constitute Debt Obligations?

The debentures issued by the western mortgage companies in the 1880s differed from the companies' sales of individual mortgage loans inasmuch as the sales of individual loans represented sales of assets, with the loans assigned to the individual investors in the local land records. By contrast, the debentures represented debt obligations of the mortgage companies that were collateralized by mortgages deposited with the companies' trustees. The New York State Superintendent of Banks characterized the debt obligation status of the securities issued by the western farm mortgage companies as follows:

The form of securities most largely dealt in by the investment companies are mortgages secured by real estate—either farm or city property—and debentures. These latter are securities issued by the company as a personal, direct obligation, the payment of which is secured by depositing with certain persons or corporations, securities in trust, with power in

the trustee to sell and convert these securities into money and apply the money towards the payment of the debentures or the interest thereon, whenever the company fails to meet such payments when due.

(Preston, 1891, p. 9)

The debt obligation status of the securities meant that if the companies failed to make timely payments on the debentures, the creditors could lay claim to the assets of the mortgage companies.

In contrast to the clarity around the debt obligation status of the 1880s debentures, ambiguity surrounded the contractual form of the mortgage participation certificates in the 1900s. The language describing the rights of mortgage certificate investors lacked clarity and varied across the certificates. The ambiguity allowed room for categorizing the certificates either as a sale of assets entitling the investors to fractional ownership of the underlying mortgages, or a secured debt obligation of the issuers (*Columbia Law Review*, 1934, pp. 675–676).

The ambiguity in the 1920s certificates shaped the lessons drawn by the contemporaries from the market's meltdown. Specifically, one lesson drawn from the aftermath of the issuers' bankruptcies was about the undesirability of selling mortgage participations no matter their contractual form. A New York State legislature committee charged with proposing reform of the mortgage banking system attributed the system's failure to the dangers associated with the practice of selling MBS:

The public investments issued by the mortgage bank are to be general obligations of the bank and not shares or participations in individual mortgages or groups thereof. This is a fundamental requirement of future mortgage financing. *Our experience has shown that the practice of selling shares or participations in individual mortgages or groups, or of selling bonds secured by an individual mortgage or group, is unfair, calculated to deceive the public, and fraught with great danger* [emphasis added].

(Barker, 1936, p. 6)

This conclusion points to two important commonalities between the first two MBS markets. First, despite the differences in the debt obligation status, in both markets the securities' issuers exercised control over the composition of the securities' collateral after the security issuance. This control meant that investors could not be certain that the mortgages that originally constituted the collateral of the securities they invested in would remain in the collateral. Second, like the sale of debentures in the first MBS market, the sale of certificates in the second market allowed the issuers to keep the profits from selling the securities on the balance sheet without acknowledging the costs associated with meeting the guarantees of loan performance.

The MBS issuers in the three markets varied in their approaches to whether the securities constituted a debt obligation of the issuers. The western mortgage companies of the 1880s issued debentures—securities that represented obligations of their issuers backed by mortgage collateral. The mortgage insurance companies

of the 1900s issued certificates that represented participations in mortgage pools, leaving ambiguous the exact status of the security on the debt obligation dimension. The distinction between whether the securities represented a sale of assets or an obligation of the issuer preoccupied the MBS designers in the third market for almost 18 years, between 1968 and 1986. In all three markets, the distinction was important to the financial health of the securities' issuers who sought to remove the risks associated with the mortgages from their balance sheets.

Role of Repayment Guarantees in Market Development

In all three markets, the issuers sought to assuage investor concerns about risks associated with the mortgages in the securities' collateral by guaranteeing the performance of the loans in the securities' collateral. These guarantees of loan performance played an important role in the investors' decision-making. Promoters of western mortgage investments conceptualized investors' due diligence research as needing to focus on picking the right company and then trusting its guarantees:

The investor has only one duty to perform, withal a very important one,—viz., to *determine once for all whether the company he is dealing with, by virtue of the character and ability of its officials, its established methods of business, the amount of its capital, and the availability of its assets, is able to give him a good and sufficient guarantee.* That there are such companies, is shown by the fact that hundreds of millions of dollars have been thus invested, without the loss of a dollar of principal or interest; and each year the examinations of these companies made by the Bank Commissioners of the Eastern States, whose savings-banks buy so largely of these loans, and which therefore insist upon these examinations, make this sole duty of the investor a comparatively simple and easy one [emphasis added].

(McGeorge, 1890, p. 431)

In the 1900s, MBS issuers framed the guarantees they offered as a form of insurance. Most of the companies offering the participation certificates were originally licensed to sell title insurance and were regulated by the New York State Department of Insurance. However, the market's crash made obvious that the guarantees offered by the companies had little in common with other lines of insurance:

Guaranteeing payment of principal of a mortgage has been enormously profitable, but *although it was considered to be a form of insurance, there was absolutely no actuarial basis upon which it rested.* There were no restrictions whatsoever in the law limiting the size of the guarantee of an individual mortgage in relation to total capital funds, and there was no requirement that total guarantee liability should be restricted to a certain number of times capital funds [emphasis added].

(Weil, 1933, p. RE1)

The guarantee of interest and principal payments offered by the issuers tied the mortgage participation certificates to the financials of the mortgage insurance companies.

Although some experts were aware of the guarantees' limitations, the guarantees also played an important role in shaping the investor decision-making in the 1920s. The degree of disillusionment with the promise of guarantees attests to the public's prior faith in the instruments:

The public has learned that there is no magic in the fact that a mortgage certificate was guaranteed. *There was an extraordinary amount of misunderstanding of the value as an investment of a guaranteed mortgage participation certificate.* There were many who believed that it was a legal investment for trust funds and infants because it was guaranteed. This is not the law. Such an investment was legal because it represented a first mortgage on unencumbered real estate and not because it was guaranteed. An unguaranteed mortgage certificate would be legal for trust funds provided the mortgage did not exceed two-thirds of the appraised value of the mortgaged property. A mortgage certificate would not be legal for trust funds, even though guaranteed, if the mortgage exceeded two-thirds of the appraised value of the mortgaged property [emphasis added].

(Weil, 1933, p. RE1)

As this quote suggests, investors erroneously saw the guarantee as a prerequisite to making certificates a legal form of investment in the second MBS market.

Mortgage repayment guarantees were an important factor for attracting investors to MBS in the third market. The discussion of security design prior to the securities' issuance recognized the value of a government guarantee: "A Federally guaranteed debenture would overcome all of these problems and prove attractive to all lenders" (President's Committee on Urban Housing, 1968, p. 132). The MBS market that emerged in the 1970s originally relied on the default risk guarantees offered by the federal government. Chuck Graham, the treasurer of California employees' pension funds, described his interest in GNMA pass-through certificates as follows: "I like them because they have virtually a double guarantee. The underlying mortgages are guaranteed by the FHA or the VA, and interest and principal are guaranteed by the U.S. government" (Clowes, 1978, p. 19).

While the federal government offered investors protection from default risk of the securities in the third market, there was no protection from prepayment risk—the risk that mortgage borrowers would repay their loans before maturity date.¹¹ To help protect investors from this risk, the market participants developed tranching—a set of tools that directed the cash flows from the collateral to different groups of investors. These tools took on an important role in the development of the MBS market, first by attracting bond investors' capital to MBS and then in creating the market for non-government-backed MBS in which tranching tools replaced government guarantees for default risk. The replacement of government guarantees with private guarantees played an important role in the unraveling of the third MBS market.

Failure to Recognize the Systemic Risk

Tranching tools that protected investors from risks inherent in MBS from 1980s onwards relied explicitly on risk diversification to offer investors protection first from prepayment and then from default risk. These tools failed first in 1994 and then in 2008 because they relied on the assumption that the borrower behavior was subject to idiosyncratic rather than systemic risk (Vinokurova, 2019a). Systemic risk in these markets stemmed from the development of MBS markets inducing a high degree of correlation in borrower behavior and real estate prices.

A growth in this correlation translated into systemic risks that played an important role in unraveling all three markets. The development of all three markets followed the same trajectory. First, the mortgage securities attracted capital to financing mortgages. Then, easier access to mortgage financing drove competition among lenders, leading to overlending that triggered real estate price inflation and speculation. Overlending resulted in riskier mortgages that were more sensitive to downturns in housing prices, rendering the securities less safe. Recognition of the risks inherent in the securities led to investor flight out of the markets and a downturn in the housing prices.

In all three crises, those who tried explaining what happened framed the problem as the actions of individuals rather than the system as a whole. In the first market, this problem was framed as one of reckless practices by individual companies:

Those who have investigated the matter for themselves, however, well know that the companies referred to have failed, *not because of any inherent weakness in the system itself*, but because of the reckless practices of the companies concerned, which were brought to light in the investigations which followed their default [emphasis added].

(McGeorge, 1890, pp. 432–433)

The focus on the individuals precluded market observers from diagnosing the systematic weaknesses and the potential for systematic remedies to such weaknesses.

The extent to which contemporaries of the first market pointed out the needed improvements, they saw listing the securities on an exchange as a way to draw the scrutiny of institutional investors:

And the bonds themselves are not as attractive as investments as the railroad loans, because they are not listed, and so are not readily convertible into cash. Nor can it be doubted that the listing of the bonds has the effect of directing the attention of the bondholders to the affairs of the companies far more effectively than when each bond owner is acting singly for himself. It is after all a very simple matter to investigate the affairs of a loan company. Having a list of the borrowers from the trustees, the

examination of a few of the loans will soon show how these have been made. The expense is the only obstacle, at present, preventing the holder of one or two bonds from doing this for himself, an expense which would be nominal when many bondholders participated. The recent failure of a few well known American companies must largely be attributed to speculation by the managers, which they would never have dared to commence in case their bonds had been subject to the vigilant scrutiny of the Wall Street speculators.

(Frederiksen, 1894, pp. 216–218)

This quote suggests that the failure to list the debentures on public exchanges made the securities less attractive not just compared to individual mortgages, but also to other securities listed on the exchanges.

In discussing improvements to the safety of the system, observers of the second MBS market cited individual companies' experience of adopting safeguards in issuing guarantees as insufficient to secure the system:

A negligible number of companies pursued certain rules based on an arbitrary ratio [of outstanding mortgage guarantees to the company's capital] and not on experience. One of the largest companies limited its total guarantee liability to twenty times its capital funds, but beyond that there was no basis for the formulation of rules.

(Weil, 1933, p. RE1)

The above quote highlights the few security issuers' attempts at self-imposed safeguards in contrast to the lack of care exercised by the other MBS issuers.

As the issuance of private-label, non-government-backed securities took on a greater role in the lead-up to the 2008 crisis, the credit ratings assigned to the resultant securities created an impression of equivalence between guarantees issued by private parties and the federal government (MacKenzie, 2011). This equivalence obscured from view an important difference between the two types of guarantee: private guarantees suitable for protecting investors from idiosyncratic risk and government guarantees capable of withstanding systematic risks.

In the first two MBS markets, the mortgage companies invested the assets that were supposed to back the mortgage payment guarantees they provided in land, mortgages, and real estate, underappreciating the systemic nature of the risk associated with a downturn in real estate markets. In the third market, the risk-management models assumed low correlation in default risk among the mortgages in the pools backing the securities and across the pools (MacKenzie, 2011). In all three markets, the possibility of creating private insurance solutions for systematic risks outlived the securities in question. Rather than convincing fellow entrepreneurs of the impossibility of such insurance, 1890 saw proposals for creating an insurance company to insure these risks (*American Hebrew*, 1890) and tranching—a form of private insurance—outlived the 2008 market (Vinokurova, 2019a).

Discussion and Conclusion

The structures of the MBS securities in the three markets described above share important commonalities that point to a lack of institutional memory in the system. Despite the crises following each of the three markets, the MBS issuers persisted in their experiments with the debt obligation status of the securities, claiming that the diversification inherent in the mortgage pool lowered the risks associated with MBS, and leveraging the illusion of government supervision in issuing the securities. Furthermore, the securities' issuers persistently failed to recognize the systemic risks inherent in real estate lending and investing.

Indeed, MBS issuers' persistence in structuring the securities is remarkable, given important changes in the institutional environment that took place in the period between the disappearance of the second MBS market and the appearance of the third. These included changes in the structure of the dominant mortgage loan from a 3–5 year non-amortizing loan with semi-annual payments to a 30-year amortizing loan with monthly payments, the targeting of institutional rather than individual investors, and, finally, complete turnover of the MBS issuers—no MBS issuer was active in any two of the three markets. Arguably, these changes in the structure of the institutional environment make it difficult to make the case that the MBS markets benefited from the existence of an institutional memory.

If not institutional memory, what then explains the persistence of the MBS structures across the three generations of the securities? One important aspect of the institutional environment that remained unchanged across the three markets is the lax regulation of mortgage lending and MBS issuance in the U.S. and fragmented regulatory structure that lacked either a centralized exchange or a single regulatory authority. Management scholars have hypothesized that regulatory regimes make room for individual market participants' actions (Funk and Hirschman, 2014). The analysis above is consistent with Snowden's (1995) and Simkovic's (2013) suggestion that the regulatory regime of the U.S. MBS markets may account for the persistence of the MBS issuers' practices. The coordination mechanisms missing from the state institutional framework provided were not made up by private coordination efforts.

The absence of coordination mechanisms contributed to the crises that ended the MBS markets on two levels. On one level the lack of coordination among MBS investors hindered both the effective monitoring of the MBS issuance practices and the resolution of mortgage defaults in the collateral. In the absence of institutions facilitating collective action such as the listing of the securities on an exchange, investors across the three markets were not able to observe and put a halt to the deterioration in the lending standards. In case of security default, triggered by defaults of mortgages in the securities' collateral the existence of multiple investors in the same pool of mortgages hindered the speedy resolution of the foreclosures—a collective action problem that became more evident as the mortgage defaults multiplied. Following a period of lackluster crops western mortgagors defaulted *en masse* leading the observers of the 1880s markets to comment that “the investor under the debenture system is without speedy remedy. He is part of a series. He

cannot move independently” (Gleed, 1890, p. 105). To address the collective action problem associated with the widespread foreclosures in the aftermath of the Great Depression, the New York State Superintendent of Insurance formed the Mortgage Protection Corporation that coordinated the actions of the multiple holders of the certificates (Weil, 1933). Such collective action problems were also evident in the aftermath of the 2008 crisis, when loan securitization status affected homeowners’ ability to obtain mortgage modifications.

The second level of a missing coordination mechanism is between the primary (mortgage lending) and secondary (MBS issuance) markets. Coordination at this level could have informed the investors about how changes in the secondary market affect the practices of the participants in the first. Investing in MBS required investor faith in the MBS issuers’ efficacy of managing the risks associated with mortgage lending. The MBS issuers’ efforts to manage the risk relied on risk diversification stemming from the securities collateral consisting of multiple loans. As more investor capital poured into MBS, this influx increased the availability of lending capital. The availability of lending capital led to real estate speculation and riskier lending—thus, lowering the overall quality of borrowers and raising the level of systemic risk in the system. The mortgage portfolio diversification that was effective in insuring against the idiosyncratic risk of the individual mortgages could not overcome this rise in systemic risk. In all three markets, the MBS market development drew capital to the mortgage market, leading to changes in the lending standards, overlending, and the subsequent crises.

Exacerbating the absence of coordination mechanisms in the system was MBS issuers’ reliance on the illusion of government oversight. Investors interpreted the issuers’ claims of government oversight as stringent governmental controls that could have provided the coordination mechanisms described above. In the western farm mortgage market, the banking regulators of eastern states started auditing the western mortgage companies soliciting investors in their state. While the regulators disavowed any knowledge of the quality of the underlying mortgages, their work lent credibility to the securities issuers by creating the impression of government scrutiny of the mortgage companies’ operation. In the second MBS market, the title insurance companies operated under the supervision of the New York State Department of Insurance. Despite the companies advertising such supervision, the insurance department lacked the staff and the expertise necessary to conduct regular audits of the companies. The participation of (quasi-)government agencies including Ginnie Mae, Fannie Mae, and Freddie Mac in the third market similarly translated the agencies’ status into an illusion of government oversight.

Implications for the Study of Financial Markets

The analysis of the similarities and differences across the three MBS markets allows insight into the role played by the failure of institutional memory in the development of financial instruments. The three markets share institutional similarities—most importantly a collective action problem facing MBS investors due to a lack

of a centralized exchange for trading the securities, lack of transparency of the securities' collateral, and limited government oversight over security issuance. These similarities led to the persistence of the security structures this chapter documents. Moreover, it is not clear that the market participants in the second and third markets were aware of the prior experiments with securitization. The market participants in the third market who knew about the drastic reductions in house price values during the Great Depression framed the recurrence of such reductions as an unlikely event. The decision-making of the firms and regulators devising the new securities for the third market reflects no awareness of the MBS issuance in the 1880s or 1920s (Ross, 1989).

One of the issues raised by this account is whether investing in MBS offers better returns compared to investing in either individual mortgage loans or other, publicly listed securities. In the first two markets, more sophisticated investors opted for holding individual loans instead of the securities on offer (Alger, 1934; Frederiksen, 1894). In the third market, a combination of federal guarantees and risk management tools grounded in risk diversification attracted institutional investors to the market. The presence of these sophisticated investors did not resolve the issues that plagued the earlier MBS markets.

My analysis identifies two challenges faced by market participants in learning from the crises: 1) drawing the right lessons and 2) remembering the lessons drawn. The re-emergence of the MBS markets in the U.S. is symptomatic of failure on both fronts. All three markets suffered from a lack of regulatory supervision, lack of transparency into the security collateral, and lack of appreciation for the systemic risks inherent in MBS issuance. The former two issues were evident to market contemporaries in the first two markets, but not remedied in the third market. The third issue continues to affect the market.

This analysis suggests that memory plays a crucial role for preventing the recurrence of financial crises. Even if the right lessons are drawn, without memory for what lessons triggered institutional change in the crisis aftermath, this change can be reversed, allowing for crisis recurrence. The repeal of the Glass-Steagall Act in 1999 is an example of what happens when the passage of time erases the memory of why institutional change was necessary. Persistence of institutional change in the aftermath of financial crises requires a combination of institutional change and a means to preserve the memory of the connection between the change and the events the change is meant to forestall.

The case examined in this chapter illustrates the relationship between institutional memory and financial crises. Failure to identify, remedy, and memorialize the connection between the financial instrument and its consequences in the aftermath of a crisis enables institutional forgetting and, consequently, crisis recurrence. The conditions that enable the re-emergence of financial instruments and the associated crises are the persistence of the institutional framework and a failure of institutional memory for the instruments' prior performance. Absent changes in the institutional setting, the re-emergence of securities that failed in previous markets will produce the same failure. The failure of memory allows the financial instruments

that failed in the past to reemerge. Treating the re-emergence of these instruments as innovations forgoes and forecloses learning from the past, setting the stage for the next crisis.

Finally, this chapter has implications for the role played by anthropology and history in the study of management. Specifically, historical analysis offers lessons with the benefit of hindsight, uncovering the process of institutional evolution over time. The cyclical nature of the management field's attention to collecting facts and the extent to which crises trigger such attention is part of a self-reinforcing mechanism. Forgetting the events of the past results in re-emergence of practices that brought about past crises. An appreciation for the role of history could forestall such crises. Arguably, a similar dynamic holds for the study of anthropology. Abstractions from a nuanced understanding of market participants' beliefs and practices embedded in the theories of human behavior lead to practices that rely on assumptions hidden in the theories. Crises highlight these limitations, reestablishing the value of anthropological research in the management field.

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Notes

- 1 The process of abstraction from facts to stylized facts has been documented in sociology (Mizruchi and Fein, 1999), economics (Coase, 2000), and the work of management consultants (McKenna, 2009). In the case of history and anthropology, such abstraction contributes to the erasure of an appreciation of the disciplines' contributions.
- 2 My discussion of epistemic cycles is distinct from Kuhn's (1962) analysis of paradigm shifts. Kuhn focuses on the emergence and decline of theories, while the epistemic cycles I describe focus on methods of inquiry. A change in the popularity of methods can occur without new theory emergence. For instance, the popularity of experimental methods in economics is yet to produce a new theoretical paradigm (Quiggin, 2010). My conceptualization of an epistemic cycle is closer to Fleck's (1979) idea that theories cycle from postulation to justification and back.
- 3 Snowden (1995) argued that such continuity had to do with the European governments stepping in to relieve the crises that occurred in the markets in a way that preserved the integrity of mortgage lending institutions.
- 4 These periodicals ranged from religious and regional newspapers in the earlier periods including *American Hebrew*, *Chicago Daily Tribune*, *Christian Union*, *Los Angeles Times*, *New York Evangelist*, and *Philadelphia Inquirer* to national newspapers in later periods, i.e., *New York Times*, *Wall Street Journal*, and *Washington Post*. For the third market, I supplement my analysis of the structure of the securities with reading the securities' prospectuses or offering circulars filed with the Securities and Exchange Commission (SEC) and the Federal Home Loan Bank Board (FHLBB).

- 5 In the 1880s, these included the reports of the Connecticut, Massachusetts, and New York banking regulators. In the 1900s, these were the Banking and Insurance Departments of the State of New York. In the 1970s, these included the FHLBB, SEC, the Office of the Comptroller of the Currency (OCC), the Office of Thrift Supervision (OTS), the Federal Reserve, and the Federal Deposit Insurance Corporation (FDIC), among others.
- 6 The discussion of the 1880s market draws on Frederiksen (1894), Bogue (1955) and Snowden (1995); the 1900s—Alger (1934); and 1970s onwards—my own research (Vinokurova, 2018, 2019a, 2019b).
- 7 The act reflected the challenges to the Johnson administration's issuance of participation certificates in assets of various federal agencies including Fannie Mae to get around the debt ceiling (Quinn, 2010). Members of the U.S. Congress challenged these attempts, observing that the federal government remained liable for the risks associated with the assets while raising funds more expensively than if the obligations were issued by the U.S. Treasury.
- 8 The government institutions created in the wake of the Great Depression popularized changes in the structure of residential mortgages.
- 9 Individual investors who provided 50% of the funding for residential mortgages before the Great Depression were almost entirely out of the market by the 1950s.
- 10 This inconvenience also affected investors in pay-through bonds. Once the mortgage collateral moved off the balance sheet of the issuer, the timing of the securities' repayment and, consequently, their yield, was guesswork.
- 11 This risk was less problematic for the first two MBS markets because of shorter maturities of the mortgages and the availability of prepayment penalties.

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